MACS 40550: Midterm Exercise 1

Annie Kong

April 17, 2026

Research Question

The Schelling segregation model shows that even when agents do not have an explicit preference for same-group neighbors, strong residential segregation can still emerge over time. My research question is whether life-stage-based preferences—where households with children, young adults, and elderly residents prefer different neighborhood conditions—can produce spatial segregation comparable to the identity-based segregation in the original model.

Hypothesis

My hypothesis is that life-stage preferences can produce significant spatial clustering among same-group people living in neighborhoods, even without identity-based bias. There are three types of life stages in my modification. First, family agents will cluster in high-density areas near other families because this may be close to school districts and they are more willing to live together. Second, elderly agents will settle in low-density neighborhoods because they may prefer more quiet and less crowded areas. Third, young adult agents will occupy more mixed areas because they are more flexible about their surrounding environment. The overall share of happy agents is expected to stabilize at a high level compared to the original model, though it is not necessarily to reach full convergence due to the conflicting preferences across life stages.

Description of Modification

The modification changes the decision rule from an identity-based preference to a life-stage-dependent preference. The major change is a new agent attribute 'life_stage' that is assigned randomly at initialization to one of three life stages: 'family', 'young', or 'elderly'. I also replaced the single shared desired_share_alike threshold with a unified desired_score threshold applied to three distinct scoring functions. The grid, movement to empty cell, and data collection remain unchanged. The three decision rules are as described below:

- 'Family' agents compute their score as the proportion of neighbors who are also family agents. This captures the real-life tendency that parents are more likely and willing to live near other families with similar school-age children.

- 'Young' agents apply the original Schelling decision rule — the proportion of neighbors sharing the same agent type. This group preserves a baseline comparable to the unmodified model and keeps the simulation stable.
- 'Elderly' agents prefer lower-density surroundings and quieter areas. Their score equals one minus the local occupancy rate (valid neighbors divided by maximum possible neighbors in the Moore neighborhood), so fewer neighbors yields higher satisfaction.

I added modification notes in my code to highlight the changes I made. In app.py, I also updated the color for each agent by life stage to make the clustering more visually apparent: green = family, blue = young, orange = elderly.

Conceptual Change

The original model is interpreted as the study of racial or ethnic segregation in which agents represent households of different racial or ethnic backgrounds. In this modified model, the life-stage model shifts from identity-based to needs-based. The agents represent the households at different points in the life cycle, and each has legitimate and non-prejudicial reasons for preferring certain neighborhood types. This reflects the real housing market patterns, such as school-district clustering and retirement communities. It broadens the conceptual scope of the model by asking whether the local segregation dynamic can emerge purely from lifestyle preference, rather than from any bias or discrimination. The changes demonstrate that segregation arise not only from identity-based preferences, but also from differences in functional or lifestyle needs.

Speculative Results

From the simulation results, the hypothesis is supported by the results that life-stage clustering emerges rapidly and stabilizes at a level comparable to the original model. From the dynamics, the share of happy agents rises steeply within the first 150 steps and then plateaus around 90%, without fully converging to 100%. This persistent fluctuation highlights a key difference from the original model that elderly agents continuously reassess local density as the neighbors relocate, which prevents the system from reaching a fully stable equilibrium. This phenomenon shows that the conflicting preferences across life stages keep the system in a quasi-stable rather than a fully converged state.

Spatially, the family and young agents formed clearly visible contiguous clusters on the grid. The elderly agents are dispersed in the low-density area that was left by the other two agent groups. This pattern shows that clustering with other elderly agents increases the local density and reduces their satisfaction. This contrast between clustering and dispersal is a key emergent finding. Blue (young) agents also tend to surround green (family) clusters, as both may share the same agent type under the Schelling rule. At the later stages of the simulation, only orange (elderly) agents continue to relocate, as family and young agents stabilize quickly while elderly agents remain sensitive to local density changes. This explains why the share of happy agents plateaus near 90% rather than converging to 100%.

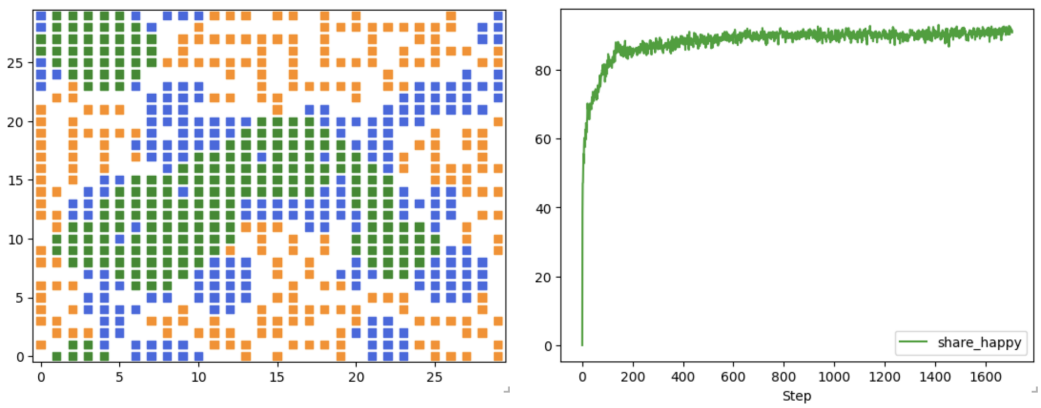

Figure 1: Spatial distribution of agents by life stage and the share of satisfied agents over time

Note

The overall modification was based on the Schelling model code provided in class. I used Mesa documentation and Claude for debugging.

Yue, X., Wang, Y., Li, W., Wu, Y., Wang, Y., Zhang, H. O., & Ma, Z. (2024). Research progress and trends in urban residential segregation. Buildings, 14(7), 1962.